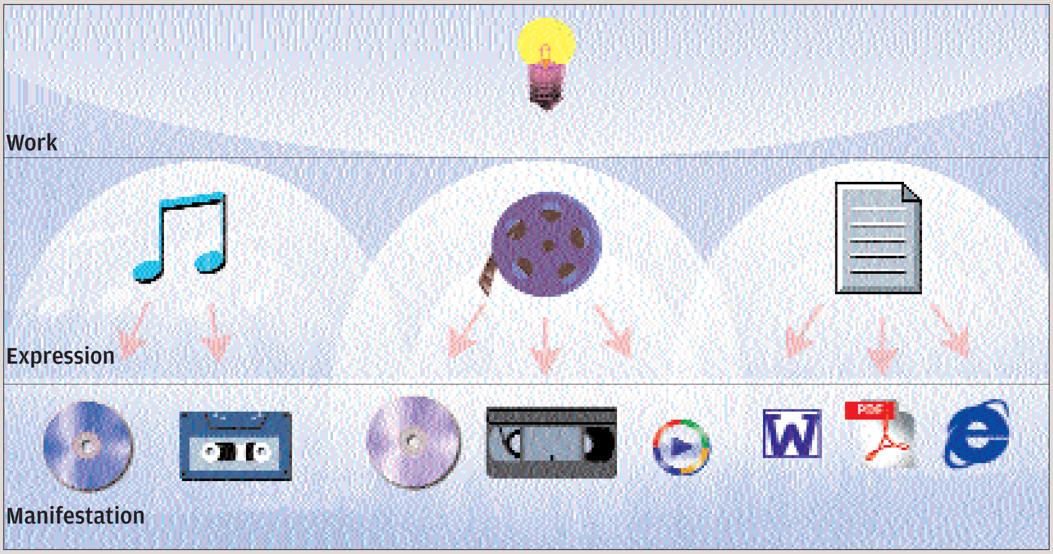
which would then be verified by their servers, and return you a final authorisation code that would unlock the software. This works better, but it's been bypassed too—just like every system that involved actually *giving* the key to the user.

When Windows XP was launched, it not only activated itself over the Internet, it also locked itself to your hardware—so your hardware ID numbers now became the key to use it. The scheme was dropped soon, because a simple hardware upgrade could put your XP out of commission. Still, the concept of using a key that users couldn't control (and therefore couldn't alter) remained, and is still evolving in the DRM techniques we see today. Windows Vista, for example, will allow you five upgrades before it recognises your PC as invalid.



The content model

While it started out as simple copy-protection, DRM has now evolved into a much more complex beast, not only defining what you can do with software and media—in an End User License Agreement, for example—but making sure that those guidelines are followed as well.

The DRM Model

Designing a DRM scheme where both users and content creators are happy, has, and will continue to, give companies sleepless nights in coming years. At the highest level, they have three aspects to consider (the jargon refers to all content—software, media, etc.—as *assets*, so we'll do the same for convenience):

1. Creation: This deals with creating protected assets and defining the rights users have over them. This is basically defining the aspects of the law under which the content is protected. Once this is done, assets are ready to be sold.

2. Management: This deals with managing asset sales—assigning licenses to users, making sure that royalties get to the creators, and so on.

3. Usage: Once you've got a DRM-ed asset, it needs to verify a lot of things. Most importantly, it needs to check whether you're authorised to use it. This aspect also involves keeping tabs on *how* the asset is being used—if there's a three-copy restriction, for example, this is validated every time the file is copied.

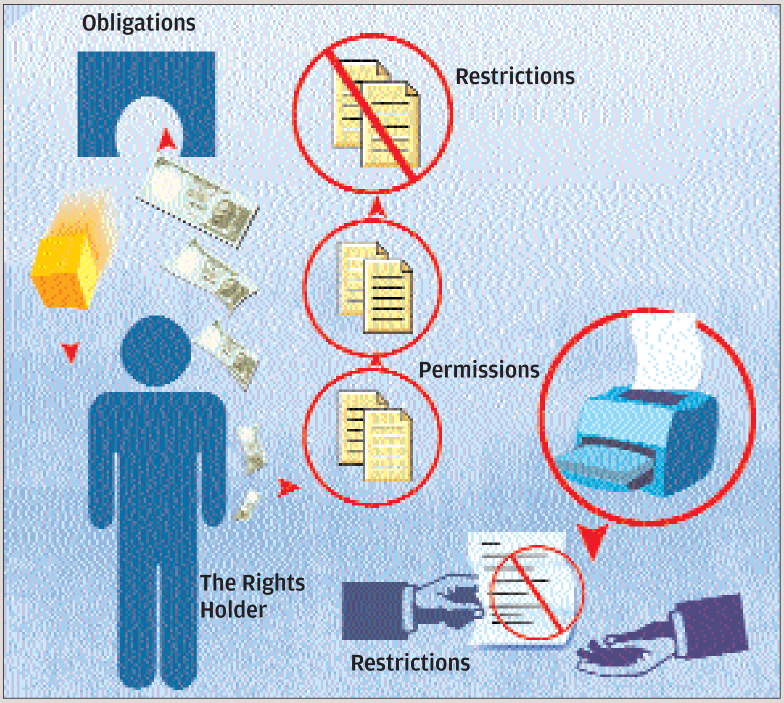
All this, of course, is a very high-level look at things. Managing protected content involves a whole bucketful of e-business models, for example, and as we'll see next, creating protected content isn't exactly a piece of cake, either.

The DRM Trinity

In the creation of protected content, we need to consider three entities—users, their rights, and the content itself. All three entities, their attributes and the relationships between them need to be expressed in terms that can eventually be translated into software. Some legal jargon follows, you have been warned.

Content is modelled as a hierarchy, starting at Work, which is the creation. The Idea for this article, while it evolved in the writer's head, is an example. The Work is then translated to Expression—the manner in which the idea takes form—in this case, the text of this article. The Expression is then dished out to the world as one or more Manifestations, which is a physical (or digital) realisation of the content.

Taking the example of the article, the original Word document, the print version you're reading now, and the PDF version you'll get in



The rights model